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ASSIGNMENT : 5

Problem Statement:
Write a program in C to perform basic matrix operations such as:
1. Addition of two matrices.
2. Saddle pt of a matrix
3. Inverse of a matrix.
4. Magic square of a matrix

Objective:
- To understand matrix representation in C using two-dimensional arrays.
- To implement various matrix operations using C programming.
- To develop logical thinking for solving matrix-based programs.

Theory: Matrix operations are widely used in engineering & scientific applications. Addition of matrices involves adding corresponding elements. A saddle pt is an element that is the minimum in its row & maximum in its column. The inverse of a matrix exists only for square matrices with non-zero determinant. A magic square is a square matrix in which the sums of elements in each row, columns, and diagonals are equal.

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ALGORITHM:

Step1: START.
Step2: Read the order of the matrix
Step3: Read elements of the matrix
Step4: Perform matrix addition.
Step5: Check for saddle pt.
Step6: Find inverse of the matrix (if determinant ≠ 0)
Step7: Check whether the matrix is a magic square.
Step8: Display results.
Step9: Stop.

C Program:

1) Addition of two matrices.

```
# include <std.io.h>
int main() {
    int a [2][2], b[2][2], sum[2][2];
    int i, j;
    printf("Enter Element of matrix A: \n");
    for (i=0; i<2; i++)
        for (j=0; j<2; j++)
            scanf("%d", &a[i][j]);
    printf("Enter matrix B: \n");
    for (i=0; i<2; i++) {
        for (j=0; j<2; j++) {
            scanf("%d", & b[i][j]);
    for (i=0; j<2; j++)
        for (j=0; j<2; j++)
            sum[i][j] = a[i][j] + b[i][j];
    printf(" Sum of matrix: \n");
```

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```
    for (i=0; i<2; i++){
        for (j=0; j<2; j++)
            printf("%d", sum[i][j]);
        printf("\n");
    }
    return 0;
}
```

2. Saddle pt of a matrix.

```
# include <stdio.h>
int main() {
    int a[3][3], j, i, k, min, col, flag=0;
    printf("Enter matrix:\n");
    for(i=0; i<3; i++)
        for (j=0; j<3; j++)
            scanf("%d", &a[i][j]);
    for(i=0; j<3; j++){
        min = a[i][0];
        col=0;
        for(j=1; j<3; j++){
            if (a[i][j] < min) {
                min = a[i][j];
                col = j;
            }
        }
        for (k=0; k<3; k++){
            if (a[k][col] > min)
                break;
        }
        if(k==3){
            printf("Saddle pt is %d \n", min);
            flag=1;
        }
}
```

```
    if (flag==0)
        printf("No saddle pt\n");
    return 0;
}
```

3) Inverse of 2x2 matrix

```
# include <stdio.h>
int main() {
    float a, b, c, d, det;
    printf("Enter elements of matrix:\n");
    scanf("%f %f %f %f", &a, &b, &c, &d);
    det = (a*d) - (b*c);
    if (det==0) {
        printf("Inverse not possible\n");
    } else {
        printf("Inverse matrix:\n");
        printf("%.2f %.2f\n", d/det, -b/det);
        printf("%.2f %.2f\n", -c/det, a/det);
    }
    return 0;
}
```

4) Magic Square check.

```
# include <stdio.h>
int main() {
    int a[3][3], i, j;
    int sum=0, row, col, d1=0, d2=0;
    int magic=1;
    printf("Enter matrix:\n");
    for(i=0; i<3; i++)
        for(j=0; j<3; j++)
            scanf("%d", &a[i][j]);
```

```
for (j=0; j<3; j++)
    sum += a[0][j];
for (i=0; j<3; i++) {
    row = col = 0;
    for (j=0; j<3; j++) {
        row += a[i][j];
        col += a[j][i];
    }
    if (row != sum || col != sum)
        magic = 0;
    d1 += a[i][j];
    d2 += a[j][i];
}
if( d1 != sum || d2 != sum)
    magic = 0;
if(magic)
    printf("Magic Square\n");
else
    printf ("Not a magic Square\n");
return 0;
}
```

Input:

Element of matrix A:
Element of matrix B:
Element of matrix:
Elements of 2x2 matrix:
Element of 3x3 square matrix:

Output:

Matrix A:

1 2

3 4

Matrix B

5 6

7 8

Sum of matrices: 6 8

10 12

Input: {saddle pt}

1 2 3

4 5 6

7 8 9

Saddle pt is 3

1 2 {Inverse matrix}

3 4

Inverse matrix:

-2.00 1.00

1.50 -0.50

{magic Square.

Conclusion: Thus, we have learned how to perform basic matrix operations using C programming & understood their practical applications.

FAQs:

1. What is a saddle point in a matrix?

A. A saddle point of a matrix is an element that

is smallest element in it's row & largest element in its column.

2. When does a matrix have an inverse?

A. A matrix has an inverse if: it is a square matrix & it's determinant is not equal to zero ($det \neq 0$).

3. What is magic square?

A. A magic square is a square matrix in which sum of elements of each row, each column & both diagonals are equal.